Frontend – Assignment 1

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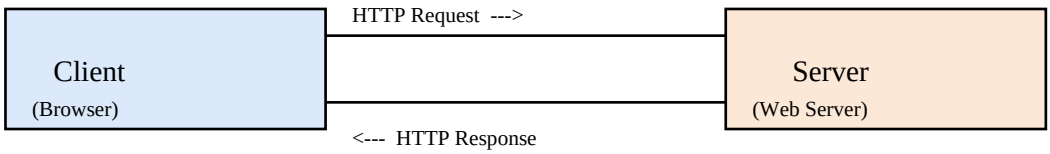
Q1. HTTP Request–Response Cycle

Whenever we open any website in our browser, there is a whole process happening in the background before the page actually shows up on our screen. This process is called the HTTP Request–Response Cycle. In simple words, it is just the client (our browser) asking the server for something, and the server giving back what was asked.



The main steps that happen in this cycle are given below:

1. User enters a URL: User types something like www.example.com in browser or clicks a link.

2. DNS Lookup: Browser first finds out the IP address of that website using DNS (Domain Name System), because computers understand IP address, not names.

3. Client sends HTTP Request: Browser now sends a request to server's IP address. This request has method (GET, POST etc), headers and sometimes data also.

4. Server processes the request: Server receives this request, understands what is being asked (maybe a HTML page, image, or some data), and processes it. It may also talk to a database if required.

5. Server sends HTTP Response: After processing, server sends back a response which has a status code (like 200 OK, 404 Not Found), headers, and the actual content (HTML/CSS/JS/images etc).

6. Browser renders the page: Browser takes this response and renders it on screen so that user can see and use the webpage.

Q2. Introduction to Web Technologies

1. HTML

HTML stands for Hyper Text Markup Language. It is not a programming language, it is a markup language which is used to give structure to a webpage, like headings, paragraphs, images, links, tables etc. Basically HTML is the skeleton of any webpage.

2. CSS

CSS stands for Cascading Style Sheets. It is used to style the HTML elements, like giving colors, fonts, spacing, layout etc. If HTML is the skeleton, CSS is like the skin and clothes which make the page look good.

3. JavaScript

JavaScript is a programming language which is used to make the webpage interactive and dynamic. Things like form validation, dropdown menus, sliders, pop ups etc are done using JavaScript. Without

JS the page will be static, means nothing will change without reloading.

4. Bootstrap

Bootstrap is a CSS framework which already has ready-made classes and components (like buttons, navbars, grids) so that developer don't have to write CSS from scratch. It also helps in making the website responsive (works well on mobile, tablet, desktop).

5. jQuery

jQuery is a JavaScript library which makes writing JS code shorter and simpler. Things which take many lines in normal JavaScript can be done in one line using jQuery, especially DOM manipulation, event handling and AJAX calls.

6. React.js

React.js is a JavaScript library made by Facebook, used for building user interfaces, mainly single page applications. It uses concept of components (reusable pieces of UI) and virtual DOM which makes the app fast because it updates only the part which changes, not the whole page.

Q3. Client-Side vs. Server-Side Technologies

Client-side technologies are the ones which run inside the user's browser, on the user's own device. Their main job is to handle what the user sees and how the page behaves in front of them. Examples: HTML, CSS, JavaScript, React.js, jQuery, Bootstrap.

Server-side technologies are the ones which run on the web server, not on user's device. They handle things like connecting to database, checking login details, processing business logic, and then sending back the final result/HTML to the client. Examples: PHP, Node.js, Python (Django/Flask), Java (Spring), ASP.NET, MySQL/MongoDB.

Basic difference: client-side code is visible to user (we can see it in browser inspect element) and runs on browser, while server-side code is hidden from user, runs on server, and only the output/result is sent to client. Also client-side is more about look and interaction of page, server-side is more about data, security and logic.

Q4. Block-Level Elements in HTML

Block-level elements are those HTML elements which always start on a new line and take up the full width available (by default), no matter how much content is actually inside them. Because of this, two block elements placed one after another will always appear on separate lines, one below the other.

Main characteristics of block-level elements:

• They start on a new line automatically.
• By default they take 100% width of their parent container.
• We can set width, height, margin, padding for them.
• They can contain other block-level elements as well as inline elements inside them.

Examples of block-level elements: ** **, **<p>**, **<h1> to <h6>**, **<ul>/<ol>/<li>**, **<table>**, **<form>**.

Q5. Semantic HTML Elements

Semantic elements are those HTML tags which clearly describe their meaning/purpose to both the browser and the developer, just by looking at the tag name. For example <header> clearly tells that this part is header of the page, we don't need to guess like we have to guess for a plain .

Some common semantic elements: <header>, <nav>, <main>, <article>, <section>, <aside>, <footer>, <figure>.

Importance of semantic elements in web development:

• It makes code more readable and easy to understand for other developers.

• It helps search engines (SEO) understand the structure and important content of page better.

• It improves accessibility, screen readers (used by blind/visually impaired users) can navigate the page properly using these tags.

• It reduces unnecessary use of everywhere, so code looks more organized.

Q6. Creating a Table Using HTML

HTML Code:

```
<table border="1">
  <tr>
    <th>Sr. No.</th>
    <th>Name</th>
    <th>Designation</th>
    <th>Department</th>
  </tr>
  <tr>
    <td>1</td>
    <td>John</td>
    <td>Manager</td>
    <td>Sales</td>
  </tr>
  <tr>
    <td>2</td>
    <td>Smith</td>
    <td>Sr. Manager</td>
    <td>Marketing</td>
  </tr>
  <tr>
    <td>3</td>
    <td>Jane</td>
    <td>Manager</td>
    <td>HR</td>
  </tr>
</table>
```

Output (how it will look on browser):

Sr. No.	Name	Designation	Department
1	John	Manager	Sales
2	Smith	Sr. Manager	Marketing
3	Jane	Manager	HR

Q7. Creating a Nested List Using HTML

Here 'River' is the main item (no numbering needed, so unordered list), and Ganga, Yamuna, Brahmaputra, Narmada are sub items of it, so nested unordered list is used inside.

```
<ul>
  <li>River
```

```
    <ul>
      <li>Ganga</li>
      <li>Yamuna</li>
      <li>Brahmaputra</li>
      <li>Narmada</li>
    </ul>
  </li>
</ul>
```

Q8. Types of CSS

There are mainly 3 types of CSS which can be used to style an HTML page:

1. Inline CSS

In this, CSS is written directly inside the HTML tag using the style attribute. It is applied only to that one element.

```
<p style="color:blue; font-size:16px;">Hello World</p>
```

2. Internal CSS

In this, CSS is written inside a <style> tag in the <head> section of same HTML file. It applies to whole page.

```
<head>
  <style>
    p { color: blue; font-size: 16px; }
  </style>
</head>
```

3. External CSS

In this, CSS is written in a separate .css file and linked to HTML file using <link> tag. This is best for big projects because same CSS file can be used for multiple pages.

```
<link rel="stylesheet" href="style.css">
```

Q9. Practical Understanding

All three, HTML, CSS and JavaScript work together to build a webpage but each one has its own role:

• **HTML** gives the basic structure/content of the page, like what elements are there (heading, paragraph, image, button etc). It is like the skeleton of body.

• **CSS** is responsible for how these elements look, like colors, fonts, spacing, layout, alignment. It is like skin, clothes and styling of the body.

• **JavaScript** gives behaviour and interactivity to the page, like what happens on button click, form validation, showing/hiding elements, fetching data without reload etc. It is like the brain and muscles which make the body move and react.

So basically HTML = structure, CSS = presentation, JavaScript = behaviour, and combining all three properly is what makes a complete, working, good looking webpage.

Q10. Short Answer

a. Role of Bootstrap in web development

Bootstrap saves a lot of time because it gives pre-built CSS classes and JS components (grid system, navbar, buttons, forms, cards etc) which we can directly use. It also makes the website responsive automatically, so it will adjust properly on mobile, tablet and desktop without writing separate media

queries every time.

b. What is jQuery and why it was widely used

jQuery is a lightweight JavaScript library which makes tasks like DOM manipulation, event handling, animations and AJAX requests much easier and shorter to write. It was widely used earlier because it also handled cross-browser issues (different browsers behaving differently), so developers didn't have to write separate code for each browser.

c. What is React.js and what problem it solves

React.js is a JavaScript library for building user interfaces using reusable components. It solves the problem of slow and messy DOM updates in large applications by using Virtual DOM, which means it only updates the part of the actual page which has changed, instead of reloading/re-rendering the whole page. This makes apps fast and also makes code more organized and reusable through components.